

# Qatar-4b

R-2664

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-4\_181113 · created 2026-08-30T00:38:46+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458435.74477 +/- 0.00066 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.0 +/- 0.037                     |
| Transit depth (Rp/Rs)^2                           | 7.5e-06 +/- 0.002 [%]             |
| Orbital Inclination (inc)                         | 89.44 +/- 2.18                    |
| Airmass coefficient 1 (a1)                        | 0.772 +/- 0.013                   |
| Airmass coefficient 2 (a2)                        | 0.219 +/- 0.054                   |
| Scatter in the residuals of the lightcurve fit is | 1.66 %                            |
| Best Comparison Star                              | #3 - [222, 326]                   |
| Optimal Method                                    | PSF photometry                    |
| Transit Duration (day)                            | 0.0753 +/- 0.0035                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                        |
|---------------------------|----------------------------------------|
| Rp/R* fit vs published    | 0.0 ± 0.037 vs 0.138 (deviation 3.73σ) |
| Duration fit vs published | 0.0753 d vs 0.0892 d (deviation 3.96σ) |
| Mid-time O-C              | -1.8 min                               |
| Depth SNR                 | 0.0                                    |
| Residual scatter          | 1.66 %                                 |

### Light curve

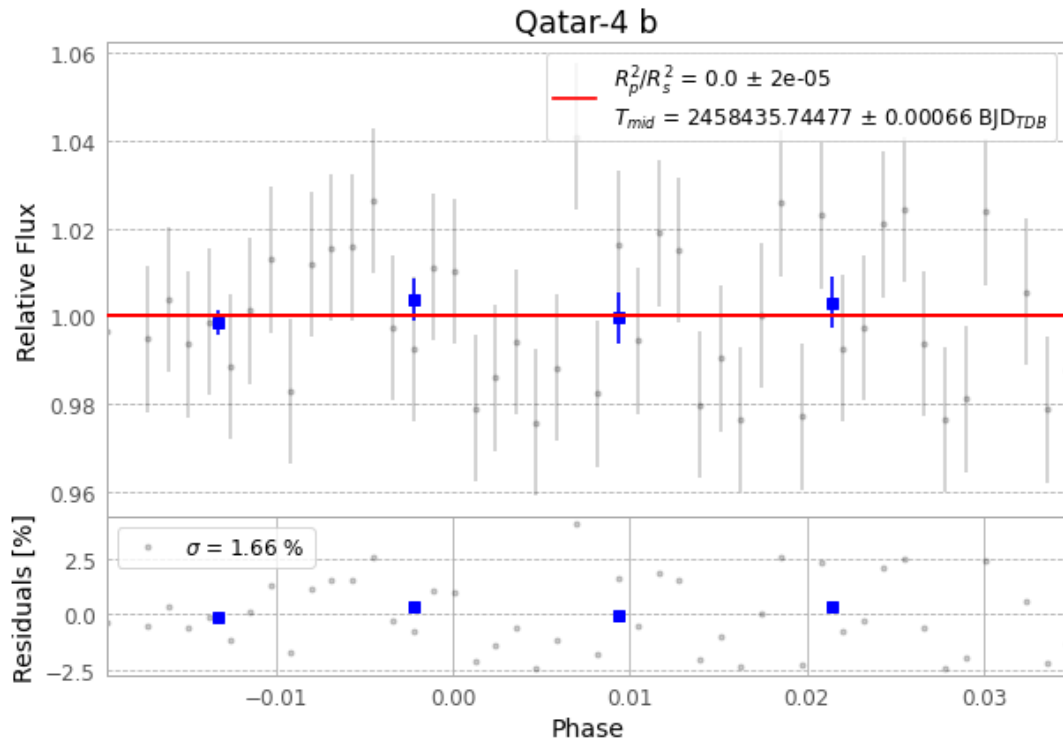


Figure 1 — FinalLightCurve\_Qatar-4 b\_2018-11-12.png (SHA-256 15f625010fecee3b...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 335], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 ebdb425c218c77f5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

47 FITS frames (31 MB), Qatar-4181113045412.FITS → Qatar-4181113071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-181113.log (d61127b20f38...), FinalLightCurve\_Qatar-4 b\_2018-11-12.png (15f625010fec...), FinalLightCurve\_Qatar-4 b\_2018-11-12.csv (5635e24977bb...)

## Compute resources

Wall time 102.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:38Z.